

HINDMARSH ISLAND, AustraliaWMO: **946770**Lat: **35.5194S**Lon: **138.8178E**Elev: **10**StdP: **101.20**Time Zone: **9.50 (AUA)**Period: **03-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.3	6.4	-1.0	3.5	23.1	0.7	4.0	18.0	14.8	13.0	13.6	13.0	3.7	10	0.620

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.9	35.1	18.2	32.2	17.8	29.1	17.3	20.7	28.0	19.9	26.6	19.1	25.4	7.1	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.6	13.5	23.2	17.7	12.7	22.5	17.0	12.1	21.7	60.0	28.3	57.0	26.5	54.5	25.5	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
13.4	12.2	11.1	DB	2.9	41.0	0.9	2.5	2.2	42.8	1.7	44.3	1.2	45.7	0.5	47.5	
			WB	1.9	22.7	0.8	0.7	1.3	23.2	0.8	23.6	0.3	24.0	-0.3	24.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.0	20.3	19.8	19.0	17.0	14.4	12.1	11.6	12.2	13.6	15.7	17.6	19.1
	DBStd	4.19	3.49	3.09	3.26	2.82	2.10	1.80	1.76	2.04	2.48	3.34	3.63	3.50
	HDD10.0	11	0	0	0	0	0	3	5	2	0	0	0	0
	HDD18.3	1146	9	12	27	60	123	187	208	192	144	98	58	27
	CDD10.0	2207	319	274	279	209	137	66	55	69	108	178	228	283
	CDD18.3	299	71	53	48	18	2	0	0	3	18	36	51	
	CDH23.3	2546	581	333	366	149	12	0	0	6	34	238	353	474
CDH26.7	1202	310	159	168	52	1	0	0	1	4	96	165	246	
Wind	WSAvg	5.9	6.6	6.4	5.9	5.2	5.1	5.0	5.6	5.8	5.9	6.1	6.4	6.4
Precipitation	PrecAvg	463	19	20	19	37	49	60	64	62	48	40	27	21
	PrecMax	682	81	107	81	107	109	175	153	117	113	102	109	69
	PrecMin	272	1	0	0	1	11	18	17	23	15	5	3	3
	PrecStd	82	19	25	18	29	23	32	28	21	23	25	20	15
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.6	36.3	35.7	32.1	25.2	20.0	19.6	23.7	26.9	33.3	35.5	37.9
		MCWB	19.1	19.6	18.3	16.9	14.1	13.2	11.7	13.2	14.2	16.3	17.0	18.2
	2%	DB	34.8	31.7	31.4	27.5	21.8	17.4	17.1	20.1	23.7	29.4	31.3	33.3
		MCWB	19.4	19.0	17.8	15.9	13.8	12.5	11.2	12.2	13.3	14.7	16.9	17.3
	5%	DB	29.8	27.0	27.3	24.1	19.7	16.2	15.9	17.8	20.9	25.5	27.5	28.6
		MCWB	18.7	19.0	17.6	15.3	13.7	12.1	11.2	11.3	12.6	14.1	16.9	17.5
	10%	DB	25.5	23.7	23.7	21.4	18.0	15.2	14.8	16.0	18.5	21.9	23.6	24.4
		MCWB	18.5	18.2	17.1	15.2	13.4	11.8	11.0	11.1	12.5	13.9	16.3	17.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.9	21.6	21.1	18.5	16.1	14.7	13.5	14.4	15.6	17.5	20.6	21.1
		MCDB	30.2	29.6	27.5	24.6	20.5	17.5	16.4	20.2	22.7	26.4	27.6	28.8
	2%	WB	20.8	20.3	19.8	17.5	15.2	13.5	12.6	13.2	14.5	16.3	18.8	19.6
		MCDB	28.9	26.6	26.2	22.9	18.9	16.2	15.3	17.8	20.6	24.1	26.3	26.2
	5%	WB	19.8	19.5	18.7	16.7	14.5	12.8	12.0	12.4	13.6	15.3	17.7	18.6
		MCDB	27.0	25.1	24.5	21.4	18.0	15.5	14.7	16.4	18.9	22.0	24.4	25.1
	10%	WB	18.9	18.8	17.9	16.0	13.9	12.2	11.4	11.6	12.9	14.5	16.7	17.7
		MCDB	25.3	23.7	22.7	20.2	17.3	14.7	14.2	15.2	17.5	20.4	22.6	23.6
Mean Daily Temperature Range	5% DB	MDBR	7.9	7.1	7.7	8.1	6.6	6.1	6.0	6.9	8.0	9.0	8.4	8.4
		MCDBR	17.5	14.8	15.7	13.7	10.0	7.8	7.8	11.0	13.6	17.1	16.6	17.7
		MCWBR	5.8	5.3	5.6	5.5	4.8	4.7	4.5	5.4	5.9	6.2	5.6	5.7
	5% WB	MCDBR	12.7	10.7	10.7	9.9	7.5	6.2	6.1	8.8	10.7	12.6	12.8	12.2
		MCWBR	5.3	4.7	4.9	5.1	4.3	4.2	4.1	5.0	5.4	5.6	5.4	5.1
Clear-Sky Solar Irradiance	taub	0.345	0.343	0.326	0.317	0.310	0.301	0.301	0.315	0.333	0.348	0.347	0.339	
	taud	2.534	2.552	2.594	2.608	2.609	2.615	2.607	2.552	2.494	2.454	2.489	2.534	
	Ebn at Noon	991	971	951	902	853	834	854	890	927	954	982	1000	
	Edh at Noon	110	104	92	80	71	66	70	83	100	113	114	111	
All-Sky Solar Radiation	RadAvg	7.66	6.66	5.39	3.88	2.63	2.18	2.36	3.28	4.51	5.88	6.98	7.55	
	RadStd	0.39	0.34	0.23	0.23	0.11	0.14	0.11	0.19	0.35	0.36	0.30	0.30	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (23 neighbors)	+0.42	N/A	N/A	N/A	N/A	N/A	N/A	-99	+158	+66